

Loss Function Optimization for TCN-based Model in *Surgical Phase Recognition*

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Motivation

Classification of surgical phases from video data has lately been a subject of attention in the medical field. The continuous rise of new ideas and high-performing Deep Learning models has aided that cause. Most of the model architectures use the standard BCE(Binary Cross-Entropy) or CE loss for multi-class classification of surgical phases. In this work, we explore the potential benefits of a custom, augmented loss function, and report the final metrics of the tested model, in comparison to its original version. To achieve that, We used an architecture named TeCNO- Surgical Phase Recognition with Multi-Stage Temporal Convolutional Networks[1], and the cholec80 public dataset which consists of 80 videos of cholecystectomy surgeries performed by 13 surgeons.

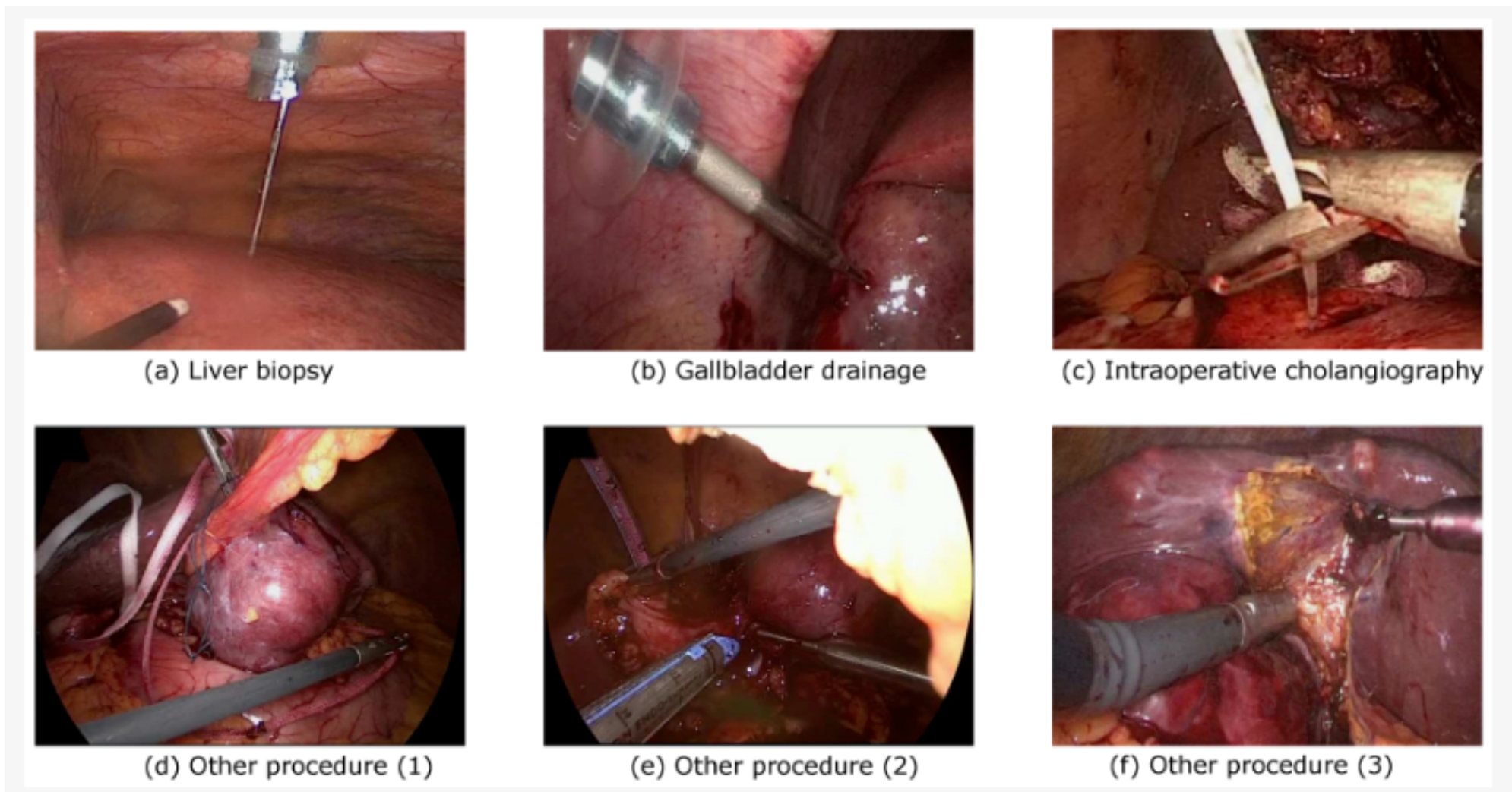


Figure 1. A sample from Cholec80 dataset

Methods

The TeCNO model can be viewed in Figure 2. TeCNO adapts the MS-TCN architecture for causal and online surgical phase recognition.

The idea is to change the loss function $\mathcal{L}_{C2}$ that is generated from the predictions of the second stage, in order to improve the model's metrics- *Accuracy*, *Precision* and *Recall*. We try several different methods to achieve this :

Multi-task loss : Use tool predictions alongside the phase predictions in order to enhance the model's capabilities, and add extra BCE with logits loss for tools.

Adjacency matrix loss : Penalize transitions that do not make logical sense in the context of the surgery.

Smoothness loss : Penalize sudden changes in the probabilities that are outputted for each phase.

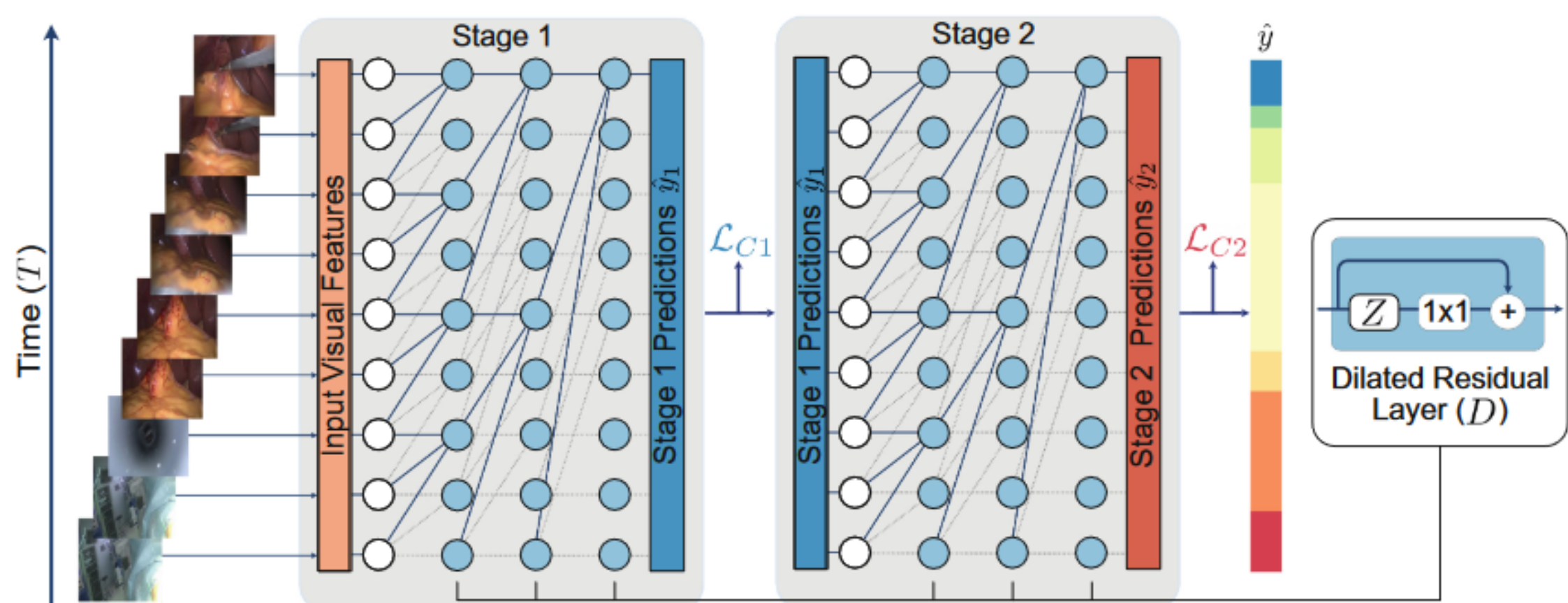


Figure 2. TeCNO Structure

Multi-task loss

The first idea we experimented on is to train the feature extractor differently so that Tool predictions are also generated and used as inputs to the TCN stages. In addition, we add an extra head in the output to allow for tool predictions from the 2nd stage TCN, as shown below :

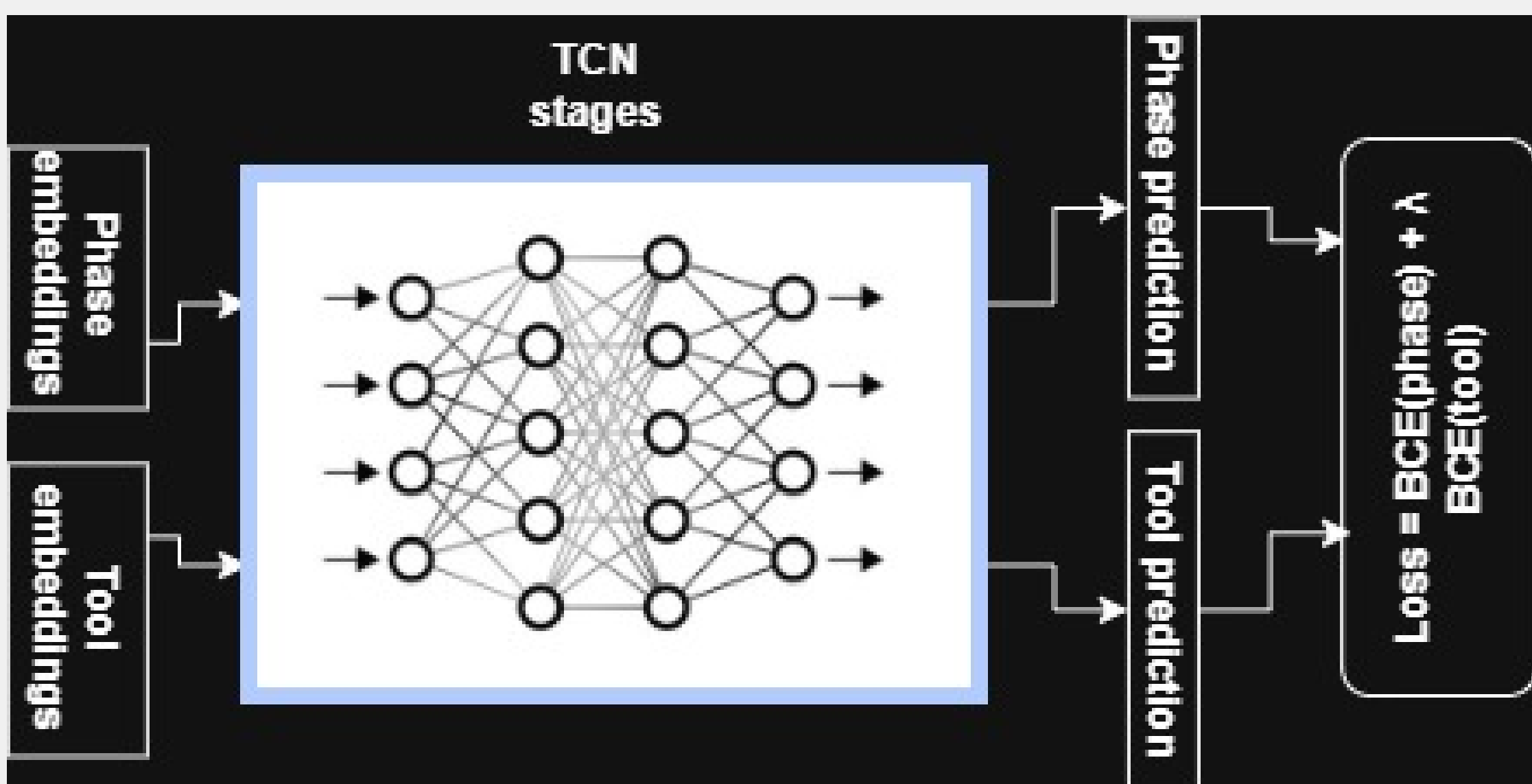


Figure 3. Multi-task loss

Then we implement the custom tool loss as :

$$L_{BCE} = -[y \cdot \log(\sigma(x)) + (1 - y) \cdot \log(1 - \sigma(x))]$$

where $\sigma(x)$ is the sigmoid function used to transform logits to probabilities, and y are the true labels. The final, augmented loss becomes :

$$L_{total} = L_{CE(phases)} + \lambda \cdot L_{BCE(tools)} = -\sum_{c=1}^C w_c \cdot y_c \cdot \log(\hat{y}_c) - \lambda \cdot [y \cdot \log(\sigma(x)) + (1 - y) \cdot \log(1 - \sigma(x))]$$

where the phases CE loss is a weighted average with frequency normalization, since some phases appear more often than others.

Transition (Adjacency matrix) Loss

Transition loss was implemented to enforce temporal consistency and penalize biologically implausible phase transitions during surgical workflow modeling.

The penalty between two consecutive time steps t and $t + 1$ is computed as:

$$\text{Penalty}_t = \mathbf{P}_t^\top \mathbf{M} \mathbf{P}_{t+1}$$

where $\mathbf{P}_t \in \mathbb{R}^C$ is the predicted softmax probabilities at time t , $\mathbf{A}$ is the binary matrix indicating allowed transitions, $\mathbf{D} \in \mathbb{R}^{C \times C}$ is the distance matrix where $D_{ij} = |i - j|$ and $\mathbf{M} = \mathbf{D} \odot (1 - \mathbf{A})$ is the penalty matrix

The loss enforces a structured prior over model outputs, so in fact we get that invalid transitions are penalized; temporal smoothness is encouraged and learning is guided using biologically-informed constraints.

Smoothness Loss

Considering that surgical phases change relatively slowly and consistently in videos, a smoothness loss was implemented. This kind of loss penalizes noisy fluctuations, enforcing smoother, more reliable transitions.

It is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{smooth}$$

where $\mathcal{L}_{CE}$ is the classification loss, computed using cross-entropy and $\mathcal{L}_{smooth}$ is the smoothness loss, which penalizes abrupt or incorrect changes in predicted phase labels, and λ as a weighting factor that balances the contribution of the smoothness loss.

Condition	Prediction Change	Penalty
Ground-truth phase does not change	Changes to wrong class	Penalized
	Changes to correct class	Not penalized
Ground-truth phase changes	Any (change or no change)	Not penalized

Table 1. Penalty rules based on prediction change and ground-truth phase transition.

Experiments

Some experiments were done aiming to prove that the custom loss functions caused an improvement in three metric scores: Accuracy, Precision and Recall. Macro-averaging of the metric scores was used, which means we compute the metrics independently for each class, then take the unweighted average across classes.

Table 2. Comparison of Multi-task loss with basic loss (mean and std)

Method	Accuracy	Precision	Recall
Multi-task loss	0.855 $\pm$ 0.012	0.783 $\pm$ 0.020	0.830 $\pm$ 0.006
Basic Loss	0.851 $\pm$ 0.014	0.774 $\pm$ 0.022	0.827 $\pm$ 0.009
Improvement(%)	0.37	0.95	0.24

Table 3. Comparison of Smoothness loss with basic loss (mean and std)

Method	Accuracy	Precision	Recall
Smoothness Loss	0.891 $\pm$ 0.006	0.797 $\pm$ 0.009	0.824 $\pm$ 0.009
Basic Loss	0.885 $\pm$ 0.008	0.796 $\pm$ 0.012	0.822 $\pm$ 0.009
Improvement(%)	0.69	0.05	0.27

Table 4. Comparison of Transition loss with basic loss (mean and std)

Method	Accuracy	Precision	Recall
Transition Loss	0.891 $\pm$ 0.005	0.799 $\pm$ 0.008	0.818 $\pm$ 0.007
Basic Loss	0.891 $\pm$ 0.004	0.801 $\pm$ 0.009	0.818 $\pm$ 0.010
Improvement(%)	0.00	-0.28	0.00

Conclusion

Using custom loss functions to enhance the model's learning capabilities showed promising results even though the rises of metric scores were not higher than 0.5-1%. Achieving a small improvement without changing the fundamental structure of a model (e.g. TCN stages, ViT (backbone)) nor its training hyperparameters, can be a very beneficial practice especially when testing results of high-performing models have saturated after tuning their hyperparameters. Finally, it is important to note that aggressive weighing of the auxiliary loss function terms sometimes lead to the model "overfitting" the secondary objective (e.g. ensure smoothness, or correct tool prediction)- therefore a careful tuning of the regularization factors is also an important factor to be considered. That said, not all custom losses were effective; the transition loss, in particular, showed no measurable benefit. Possibly because the model already captured temporal consistency through the primary loss, or because the added constraint overconstrained the system, reducing its ability to model necessary abrupt transitions.

References

- [1] Tobias Czempel, Magdalini Paschali, Matthias Keicher, Walter Simson, Hubertus Feussner, Seong Tae Kim, and Nassir Navab. Tecno: Surgical phase recognition with multi-stage temporal convolutional networks. In *Medical Image Computing and Computer Assisted Intervention - MICCAI 2020 - 23rd International Conference, Shenzhen, China, October 4-8, 2020, Proceedings, Part III*, volume 12263 of *Lecture Notes in Computer Science*, pages 343–352. Springer, 2020.
- [2] Yueming Jin, Huaxia Li, Qi Dou, Hao Chen, Jing Qin, Chi-Wing Fu, and Pheng-Ann Heng. Multi-task recurrent convolutional network with correlation loss for surgical video analysis. *Medical image analysis*, 59:101572, 2020.
- [3] Colin Lea, Michael D Flynn, Rene Vidal, Austin Reiter, and Gregory D Hager. Temporal convolutional networks for action segmentation and detection. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 156–165, 2017.
- [4] Sanat Ramesh, Diego Dall'Alba, Cristians Gonzalez, Tong Yu, Pietro Mascagni, Didier Mutter, Jacques Marescaux, Paolo Fiorini, and Nicolas Padoy. Multi-task temporal convolutional networks for joint recognition of surgical phases and steps in gastric bypass procedures. *International journal of computer assisted radiology and surgery*, 16:1111–1119, 2021.
- [5] Andru P Twinanda, Sherif Shehata, Didier Mutter, Jacques Marescaux, Michel De Mathelin, and Nicolas Padoy. Endonet: a deep architecture for recognition tasks on laparoscopic videos. *IEEE transactions on medical imaging*, 36(1):86–97, 2016.